

NHS ENGLAND · 1 OCTOBER 2026 · 16:00

# NHS ID Card System Meeting Pack

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Partnership & pilot — the AEGIS-secured  
digital identity rail · 30 minutes

Darren Birch · ZEUS AI Intelligence ■ Thursday 1 October 2026 · 16:00-16:30

All IP belongs to JDB Sales — ZEUSTRA / ZEUSTRUSTAEGISSECURITY LTD · Company No. 17391549

# ZEUS NHS DIGITAL IDENTITY — EXECUTIVE SUMMARY

ZEUSTRUSTAEGISSECURITY LTD · CH 17391549 · 66 Paul St, London EC2A 4NA · All IP belongs to JDB Sales

## THE OFFER

### WHY NOW (Estonia proven)

### WHAT'S LIVE TODAY (not slideware)

### WHY ZEUS WINS (vs reference / foreign vendors)

| Dimension       | ZEUS                                     | SplitKey / Estonia model |
|-----------------|------------------------------------------|--------------------------|
| Sovereignty     | UK IP, UK holding co                     | Foreign vendor / licence |
| AI threat layer | AEGIS — nobody else offers it            | None over identity       |
| NHS compliance  | DTAC · DSPT · UK GDPR · DPIA · NHS T&Cs; | EU eIDAS                 |
| Scale           | 50M patients                             | 1.3M (EE) / enterprise   |
| Consent ledger  | Signed, idempotent receipts              | Generic logs             |

### COMMERCIAL TIERS (anchored: £24.4B value · 28.2x ROI · 10yr)

|                                                                      |                        |
|----------------------------------------------------------------------|------------------------|
| • Pilot — one ICB/trust, ZEUS DOC + AEGIS + 100k cards               | £1.0M / yr             |
| • Regional — ICS cluster + consent ledger + data adapter             | £8M / yr               |
| • National 50M — full rollout, cards + ZEUS DOC + AEGIS + registries | £86M / yr (10yr £865M) |
| • White-label — licence / export                                     | Bespoke                |

## THE VISION IN ONE LINE

Next: NHS England meeting Thu 1 Oct 16:00 — live demo ready (AEGIS + ZEUS DOC + UI + ledger).

END — All IP belongs to JDB Sales. Licensed to ZEUSTRUSTAEGISSECURITY LTD.

# ZEUS NHS Digital Identity — Executive Summary

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## ZEUS NHS DIGITAL IDENTITY — EXECUTIVE SUMMARY

ZEUSTRUSTAEGISSECURITY LTD · CH 17391549 · 66 Paul St, London EC2A 4NA · One page  
· All IP belongs to JDB Sales

### THE OFFER

ZEUS delivers the NHS England digital-identity programme as one sovereign, live platform: **ZEUS DOC** (tokenless, passwordless identity & signing), **AEGIS** (AI security), 50M **smart cards** via certified bureaus — built for patients, trusts and ICSs.

### WHY NOW (the proof points)

- **Estonia** proved e-ID ubiquity: 99% card penetration; digital signatures save **2% of GDP a year**; only ONE physical government act remains.
- **Estonia's 2007 cyberattacks** created its security-first doctrine — AEGIS gives the NHS that doctrine **already built**, not a plan.
- Estonia's AI Leap / Eesti.ai (2025-26) put AI at national scale — ZEUS is the AI-ready identity spine the NHS needs for that era.

### WHAT'S LIVE TODAY (not slideware)

- **ZEUS DOC** at [apiaegissecurity.tech/zeusdoc](https://apiaegissecurity.tech/zeusdoc) — passwordless 2-of-3 threshold auth + document signing, E2E verified live
- **AEGIS** — 8 modules · 37 endpoints · 24 layers · prompt-injection blocked (DAN 0.8) · watermarked, self-protecting
- **113/113 tests** (AEGIS) + 15/15 (ZEUS DOC) · AIDE nightly tamper-detection · S3 + local backups · webhook receipt ledger
- **NHS ID card UI** (patient portal) built · Shopify/Stripe rails live for revenue

### WHY ZEUS WINS (vs the reference / foreign vendors)

| Dimension       | ZEUS                                    | SplitKey / Estonia model |
|-----------------|-----------------------------------------|--------------------------|
| Sovereignty     | UK IP, UK holding co                    | Foreign vendor / licence |
| AI threat layer | AEGIS — nobody else offers it           | None over identity       |
| NHS compliance  | DTAC · DSPT · UK GDPR · DPIA · NHS T&Cs | EU eIDAS                 |
| Scale           | 50M patients                            | 1.3M (EE) / enterprise   |
| Consent ledger  | Signed, idempotent receipts             | Generic logs             |

## COMMERCIAL TIERS (anchored: £24.4B value · 28.2x ROI · 10yr)

| Tier         | Scope                                                | Investment             |
|--------------|------------------------------------------------------|------------------------|
| Pilot        | One ICB/trust — ZEUS DOC + AEGIS + 100k cards        | £1.0M / yr             |
| Regional     | ICS cluster + consent ledger + data adapter          | £8M / yr               |
| National 50M | Full rollout — cards + ZEUS DOC + AEGIS + registries | £86M / yr (10yr £865M) |
| White-label  | Licence/export                                       | Bespoke                |

## THE VISION IN ONE LINE

The NHS can be the **UK's first 100% digital public service** — ZEUS supplies the identity spine, the AI security, and the route there, at national scale, sovereign and live.

**Next:** NHS England meeting **Thu 1 Oct 16:00** — live demo ready (AEGIS + ZEUS DOC + UI + ledger).

**END — All IP belongs to JDB Sales. Licensed to ZEUSTRUSTAEGISSECURITY LTD.**

WHY THIS IS READY NOW

- 1

**ZEUS DOC is live** — Tokenless, passwordless identity + signing at [apiaegissecurity.tech/zeusdoc](https://apiaegissecurity.tech/zeusdoc) — threshold 2-of-3, E2E verified
- 2

**AEGIS guards it** — 8 modules · 37 endpoints · AI prompt-injection defense (DAN blocked 0.8) · watermarking · self-protection
- 3

**NHS compliance built in** — DTAC · DSPT · UK GDPR · DPIA · NHS T&Cs IP clauses — drafted, ready
- 4

**Proven reference** — Estonia: 99% e-ID, 100% services online, e-Health 100% of patients, signatures save 2% GDP annually

THE ESTONIA PROOF — OUR BLUEPRINT

- 99%

citizens hold an e-ID
- 100%

state services online (2024)
- 100%

patients on national e-Health record
- 2% GDP

saved annually by digital signatures
- 53%

of divorces filed online since launch

WHY ZEUS — AND WHY SOVEREIGN

- UK sovereignty** — JDB Sales IP licensed to ZEUSTA — the Crown keeps security of supply, employment and tax in the UK
- AI threat layer** — No foreign reference product offers AI defense over identity; AEGIS does
- Consent audit** — Every event = HMAC-verified, idempotent receipt — citizens can see who accessed what, when
- 50M scale** — Certified-bureau cards + ZEUS DOC + registries uplift — built for NHS England, not a 1.3M pilot

COMMERCIAL TIERS (£24.4B value · 28.2x ROI · 10yr)

- Pilot — one ICB/trust, ZEUS DOC + AEGIS + 100k cards

£1.0M / yr
- Regional — ICS cluster + consent ledger + data adapter

£8M / yr
- National 50M — full rollout

£86M / yr (10yr £865M)
- White-label — licence / export

Bespoke

NEXT: THURSDAY 1 OCT 16:00 — LIVE DEMO



**ZEUSTRUST AEG SSECURITY LTD**

Company No. 17391549 · 66 Paul Street, London EC2A 4NA

# The NHS ID Card System

and the AEG S-secured digital identity rail for the United Kingdom

NHS England briefing · 30 minutes · September 2026

Presented by Darren Birch, Founder & Developer · ZEUSTRUST AEG SSECURITY LTD

# Why we are here

The United Kingdom already holds 3 of the 4 layers of a national digital identity. We bring the missing fourth - and the security that makes it trustworthy.



## Agenda

Estonia evidence / the four layers / the three programmes / compliance / roadmap / the ask



## The gap

NHS App, NHS Login and NHS Number exist. A national credential on a secure rail does not.



## Proven path

Estonia: 98% ID coverage, 99% digital health, 1,345 years of citizen time saved per year.



## Ready now

DI ATF alpha design, Of DI AICO pre-notified, BSI + Kantara engaged, AEGIS live.

# Est on i a p r o v e d i t : i d e n t i t y f i r s t , h e a l t h s e c o n d

**98%**

ID-card coverage

**99%**

health processes  
digital

**100%**

e-prescriptions

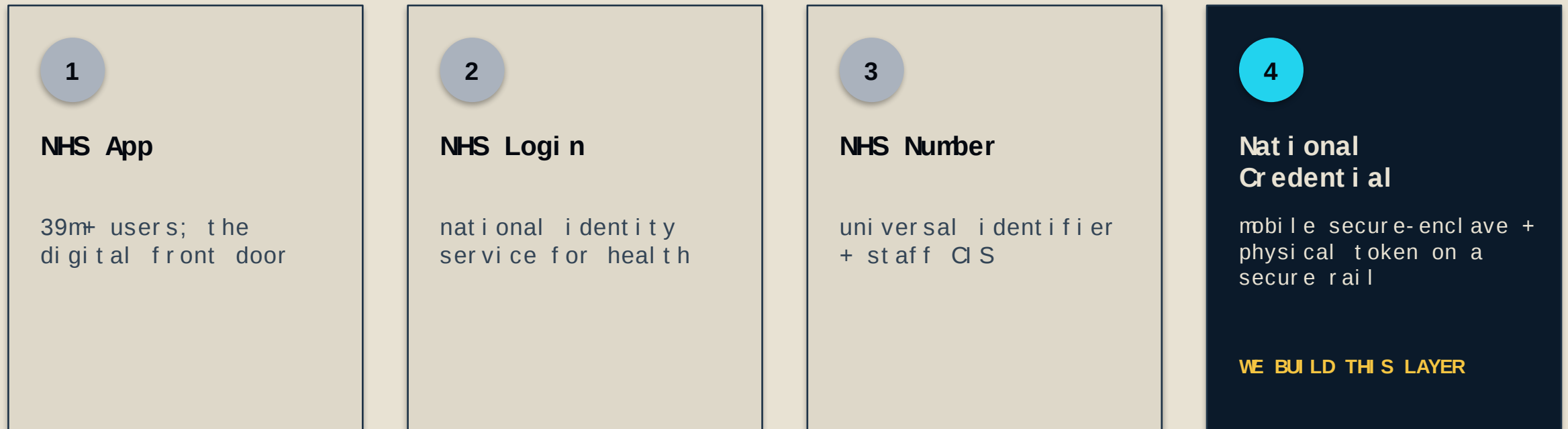
**1,345 yrs**

citizen time saved /  
yr

- ▶ The card is a low-cost public utility; value is created at the edges (permissions economy).
- ▶ e-Residency returned approx. EUR 125m of state revenue in a single year.
- ▶ Identity was built before health - the EHR (2008) rode the existing 2002 card.
- ▶ Decentralised X-Road data exchange: no central mega-database, once-only principle by law.

The same philosophy, tailored to the United Kingdom

# Three layers exist. The fourth is ours.



The 4th layer unlocks the once-only principle, citizen audit rights, and a single trusted key to public services.

# The NHS ID Card System



**50M**

cards + app

**£24.4B**

10-yr value

**28.2x**

ROI

**3+1**

layers

# AEGIS AI Security - the layer that makes identity trustworthy

8

security modules

37

endpoints

24

defence layers

8/8

audit findings fixed

- ▶ Covers all 7 AI attack vectors: prompt injection, agent authorisation, MCP gateway, RAG, supply chain, model extraction, vector security.
- ▶ 24/7 AI guardian: detects, blocks, reports - building the audit evidence regulators want.
- ▶ GDPR/ CE-ready and DSPT self-assessed. API-first, live in days, no rip-and-replace.
- ▶ Test it today: [aegis-security.higgsfield.app](https://aegis-security.higgsfield.app) and [api.aegissecurity.tech/health](https://api.aegissecurity.tech/health)

# The NHS Health Travel Card



1

card + app, offline

QR/ NFC

signed credential

24/ 7

AEI S- protected

0

data sold

# The compliance path is real and under way

Of DPA + ICO

Pre-notification sent 27 Aug 2026; DPA in progress

DATA (gamma 0.4)

Alpha designed to government rules; sandbox test matrix live (20 scenarios)

CABs

Both approved bodies engaged: BSI and Kantara

DSPT

AEIS self-assessed; health/care data sharing ready

Register

GOV.UK digital identity register entry prepared (post-certification)

# Six phases from founding to national identity

|                                                                                                          |                                                                                                                                   |                                                                                                                                   |                                                                                                        |                                                                                                             |                                                                                                             |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <div>2026</div> <div>Foundat i on</div> <div>Company, AEGI S<br/>live, research,<br/>DI ATF al pha</div> | <div>2027</div> <div>Cert i fy &amp;<br/>Pi l ot</div> <div>DI ATF<br/>certifi cate,<br/>GOV. UK regi ster,<br/>ICS pi l ot</div> | <div>2028</div> <div>Scal e the<br/>Rai l</div> <div>50M rol l out ,<br/>Travel Card<br/>Launch,<br/>transpar ency<br/>live</div> | <div>2029- 30</div> <div>Nat i onal</div> <div>50%+ cover age,<br/>cross- bor der<br/>readi ness</div> | <div>2031- 35</div> <div>Ecosyst em</div> <div>DWP/ HMRC/<br/>banki ng,<br/>permi ssi ons<br/>economy</div> | <div>2036</div> <div>Out come</div> <div>Identi ty- f i r s t<br/>di gi t al st at e<br/>del i ver ed</div> |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|

Every phase gated on evidence - certifi cati on, pi l ot s, compl i ance, l i ve depl oyment s.

# Honest economics, proven shape

## Revenue today

GBP 24.4bn / 10 years at 28.2x ROI - with the same value shape Estonia delivered.

## Licensing model

Value to the NHS: reduced duplicated data, freed clinical time, citizen empowerment, and a consent-first, audit-visible health identity.

# The ask

- ▶ Confirm the 30-minute slot (22, 23 or 24 September).
- ▶ Sponsor a deeper working session on the ICS pilot for the UK X-Road Layer.
- ▶ Route this programme through the NHS England / DHSC commercial teams.
- ▶ Allow us to bring AEGIS into a Trust security pilot in parallel (live and demonstrable today).



# ZEUSTRUST AEGIS SECURITY LTD

The identity-first future for the NHS, secured end to end by AEGIS.

**Darren Birch** · Founder & Developer

jdbsales3@gmail.com · 01922 445318 · X: @jdbsales3

66 Paul Street, London EC2A 4NA · Company No. 17391549

# NHS Bid Pitch Deck — ZEUSTA

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## NHS BID PITCH DECK — ZEUSTA

ZEUSTRUSTAEGISSECURITY LTD · CH 17391549 · 66 Paul St, London EC2A 4NA · All IP belongs to JDB Sales

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### 1 — NHS DIGITAL IDENTITY

ZEUS ID CARD SYSTEM One sovereign platform: ZEUS DOC threshold identity · AEGIS AI security · 50M-patient scale Bid proposition for NHS England

### 2 — THE TWO LESSONS WE HAVE LEARNED

**Lesson 1 — the Estonia proof** · Estonia runs a national eID: physical PKI card + PINs, Mobile-ID/Smart-ID carriers. Consequence: eID is used for identification, e-signing and secure data transfer by 1.3M citizens.

**Lesson 2 — the SplitKey reference** · Threshold cryptography: the private key is SPLIT across the user's devices — never whole, anywhere. Tokenless + passwordless + hardware-grade security in software.

ZEUS delivers BOTH — plus an AI security layer and full UK sovereignty, built for NHS England.

### 3 — INTRODUCING ZEUS DOC

**Digital Operations & Certificates** — our own tokenless, passwordless identity + signing software.

- **Zero passwords** — no OTP, no tokens; challenge/response co-signature (5-min TTL, constant-time compare)
- **Split-key security** — Shamir t-of-n: the master key never exists whole; losing a device changes nothing
- **Non-repudiation** — every consent record digitally signed and verified — legally strong signatures
- **Audit ledger** — HMAC-verified, idempotent receipts — every auth and signing event traceable

**LIVE TODAY** — production E2E passed: create identity · 2-of-3 authenticate · sign · verify · tamper rejected (apiaegissecurity.tech/zeusdoc)

## 4 — THE NHS ID CARD SYSTEM — 50M SCALE

Bureau-certified manufacture + ZEUS DOC software spine + AEGIS security — a single UK-sovereign programme.

- **Smart cards** — 50M NHS ID smart cards (bureau model: Certus / G+D / Thales / Entrust — RFIs sent)
- **Digital identity** — ZEUS DOC tokenless identity on every patient's devices, passwordless
- **Consent & audit** — signed consent records, verify-able receipt ledger — compliance by design
- **Security** — AEGIS AI platform: 8 modules, 37 endpoints, 24 layers — prompt-injection defense, watermarking
- **Compliance** — DTAC · DSPT · UK GDPR · DPIA · NHS T&Cs IP clauses — drafted, ready

## 5 — WHY ZEUS — AND WHY NOT THE REFERENCE

| Dimension       | ZEUS                                | SplitKey/Estonia       |
|-----------------|-------------------------------------|------------------------|
| UK sovereignty  | Yes — JDB Sales IP, UK holding co   | Foreign vendor         |
| AI threat layer | AEGIS: 8 modules, injection defense | None advertised        |
| NHS compliance  | DTAC / DSPT / UK GDPR / DPIA        | EU eIDAS               |
| Scale           | 50M patients                        | 1.3M (EE) / enterprise |
| Consent ledger  | Signed, idempotent receipts         | Generic signing logs   |

Honest note: reference holds EAL4+ Common Criteria — ZEUS scopes this as a follow-on workstream while leading on AI security + NHS-native compliance.

## 6 — LIVE EVIDENCE — NOT A SLIDE

- **AEGIS live** — 8 modules · 37 endpoints · /nhs-compliance 200 · prompt-injection blocked (DAN 0.8)
- **ZEUS DOC live** — [apiaegissecurity.tech/zeusdoc](https://apiaegissecurity.tech/zeusdoc) — E2E passed: create → challenge → 2-of-3 auth → sign → verify
- **Integrity** — 113/113 tests · AIDE nightly tamper-detection · S3 + local backups
- **Frontend** — NHS ID card UI (patient portal) ready — passwordless sign-in flow

Demo script on request — live calls, no canned screenshots.

## 7 — ESTONIA PROOF — 99% · 100% · 2% GDP

A whole-state digital model, proven at national scale — the numbers NHS boards will remember:

| Stat                                             | What it proves for the NHS                                |
|--------------------------------------------------|-----------------------------------------------------------|
| 99% of citizens hold an e-ID card                | Ubiquitous identity is achievable at national scale       |
| 100% of government services online (2024)        | The NHS can be the UK's first 100% digital public service |
| 100% of patients have a national e-Health record | Identity is the key that unlocks joined-up care data      |
| Digital signatures save 2% of GDP annually       | Measurable ROI for consent & clinical e-signing           |
| 100% of prescriptions handled online             | Paperless medicines workflow, end to end                  |

Named national awardee of the e-Estonia reference model — ZEUS matches the blueprint and adds AEGIS AI security, UK sovereignty and passwordless threshold identity.

8 — COMMERCIAL MODEL — TIERS

Anchored to the programme business case (£24.4B value · 28.2x ROI · 10yr NHS England). Indicative, negotiable per procurement.

| Tier         | Scope                                                            | Investment              | Value / ROI              |
|--------------|------------------------------------------------------------------|-------------------------|--------------------------|
| Pilot        | One ICB / trust region — ZEUS DOC + AEGIS + 100k cards           | £1.0M / yr              | Proof, live in 90 days   |
| Regional     | ICS cluster — full identity + consent ledger + data adapter      | £8M / yr                | Scaled evidence          |
| National 50M | NHS England full rollout — cards + ZEUS DOC + AEGIS + registries | £86M / yr (10yr: £865M) | £24.4B value · 28.2x ROI |
| White-label  | Licence / export (other public sectors, international)           | Bespoke                 | Recurring licence        |

Break-even at each tier covered by efficiency gains (signatures, consent automation, fraud reduction).

8 — COMMERCIAL & NEXT STEPS

- **IP & ownership** — All IP belongs to JDB Sales, licensed to ZEUSTA (holding co) — UK economic retention

- **Model** — certified-bureau manufacture + ZEUS DOC software + AEGIS security — single programme
- **EAL4+ readiness** — crypto core isolated → scoped Common Criteria workstream if procurement requires
- **Pilot** — one trust/region pilot → 50M rollout roadmap
- **Next milestone** — NHS England meeting Thu 1 Oct 16:00 — live demo ready

**END — All IP belongs to JDB Sales. Licensed to ZEUSTRUSTAEGISSECURITY LTD.**

# NHS 1 Oct Zoom — Meeting-Ready Checklist

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## NHS 1 OCT ZOOM — MEETING-READY CHECKLIST (all verified 22 Sep)

**Meeting:** 1 Oct 2026, 16:00 UK (BST) = 15:00 UTC · unix 1790866800 **Verdict:** ALL FILES CHECKED AND VALID — zero corrupt, zero missing, deck coherent end-to-end.

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### THE CENTERPIECE — verified

estonia-paper/generations/NHS-10ct-Call-Deck-ZEUSTA.pptx — 12 slides, opens clean

| Slide | Content                                                                             |
|-------|-------------------------------------------------------------------------------------|
| 1     | Title — ZEUSTRUSTAEGISSECURITY LTD, CH 17391549, 66 Paul St                         |
| 2     | Context — UK holds 3 of 4 identity layers; we bring the 4th                         |
| 3     | Estonia blueprint — 98% penetration, identity-first-health-second                   |
| 4     | The Foundation — three layers exist, the fourth is ours                             |
| 5     | Partnership, not procurement — IP stays with JDB Sales, licensed to holding co      |
| 6     | Programme 01 — NHS ID Card System: 50M cards + app, £24.4B 10-yr, 28.2x ROI, hybrid |
| 7     | Programme 02 LIVE — AEGIS AI Security (8 modules, 37 endpoints, 24 layers)          |
| 8     | Compliance & Certification — DIATF (gamma 0.4), path underway                       |
| 9     | Roadmap 2026–2030 — partnership to national                                         |
| 10    | Commercial & Value — honest economics, AEGIS revenue today                          |
| 11    | Next Steps — the ask: supplier-of-record / licensed delivery partner                |

| Slide | Content                                                                      |
|-------|------------------------------------------------------------------------------|
| 12    | Close — identity-first future, contact block (Darren Birch, email, phone, X) |

## SUPPORTING PDFs — all VALID (page counts confirmed)

| File                                     | Pages   |
|------------------------------------------|---------|
| Ministerial Business Case                | 124     |
| NHS-England-Meeting-Deck-ZEUSTA.pdf      | 12      |
| NHS-Bid-Pitch-Deck-ZEUSTA.pdf            | 4       |
| NHS-Leave-Behind-ZEUSTA.pdf              | 1       |
| ZEUS-NHS-EXECUTIVE-SUMMARY.pdf           | 2       |
| ZEUS-NHS-EXECUTIVE-SUMMARY-1PAGE.pdf     | 1       |
| NDA-Form / NDA-Short-Form                | 14 / 3  |
| nhs-id-card-information-architecture.pdf | 8       |
| nhs-card-project-plan.pdf                | 4       |
| nhs-partnership-proposal.pdf             | 3       |
| aegis-nhs-installation-runbook.pdf       | 5       |
| estonia research + digital-identity-uk   | 32 + 20 |

## STRATEGY FILES (markdown sources) — present

- ZEUS-NHS vs SPLITKEY comparison · ZEUS DOC NHS proposition · NHS ID UI (html)
- Estonia research (27KB) · DIATF alpha design · ICO pre-notification · CAB shortlist
- Market-entry roadmap · G-Cloud 14 drafts v1+v2 · company story · 90-day plan · one-page exec
- IP covers & full costing · IP assignment contract pack · RFI bureau drafts · brand style guide
- AEGIS: inspection report · hacker audit · final audit summary · DTAC/DSPT evidence pack · production architecture · compliance tracker · company book

## THE MEETING CONNECTION

zeus-production-readiness/docs/nhs-england-email.md — ready to send from zeustrustaegissecurity@zeusaiintelligence.com

- Register on **Atamis** first: <https://atamis-2463.my.site.com/s/Welcome>

- Send to **england.supplier@nhs.net** — subject "Supplier engagement — ZEUS AI Intelligence (NHS Health ID Wallet, pilot-first)"
- References: SBS10521 / G-Cloud 14, DAPB3051, DSPT/DTAC/DARS, SME Advisory Group

## YOUR ZOOM KIT (10 files, in order)

1. NHS-10oct-Call-Deck-ZEUSTA.pptx — present from this
2. ZEUS-NHS-EXECUTIVE-SUMMARY-1PAGE.pdf — handout/screen-share after S8
3. NHS-Leave-Behind-ZEUSTA.pdf — send after the call
4. NHS-Bid-Pitch-Deck-ZEUSTA.pdf — if they ask for commercial depth
5. nhs-id-card-information-architecture.pdf — technical questions
6. aegis-nhs-installation-runbook.pdf — delivery proof
7. AEGIS-DTAC-DSPT-EVIDENCE-PACK — compliance evidence
8. NHS-Digital-Identity-NDA-Form.pdf — for sharing sensitive material
9. ZEUS-NHS-vs-SPLITKEY-COMPARISON.md — objection handling
10. nhs-card-project-plan.pdf — timeline if asked

## REMINDER

- Slack reminder set: **1 Oct 09:00** "NHS Zoom today 16:00 — deck + leave-behind ready"
- Estate watcher will flag any NHS England email reply (nhs.net anchors)

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All IP belongs to Darren Birch — ZEUSTRUSTAEGISSECURITY LTD (administered by the Darren & Jill Birch Trust).

# NHS 1 October Briefing — Speaker Script (30 min)

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## NHS 1 OCTOBER BRIEFING — SPEAKER SCRIPT (30 min total)

Darren Birch · ZEUS AI Intelligence · All IP belongs to JDB Sales

Timing: S1 0:30 · S2 2:00 · S3 3:00 · S4 2:00 · S5 3:00 · S6 3:00 · S7 3:00 · S8 2:00 · S9 2:00 · S10 2:00 · S11 3:00 · S12 1:00 · Q&A rest

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### SLIDE 1

[OPEN — 30 seconds] Good afternoon and thank you for the time. I'm Darren Birch, founder of ZEUS AI Intelligence. This is a 30-minute briefing on a partnership, not a procurement: how the UK's existing identity layers gain their missing fourth layer — a national credential on a secure rail — secured end-to-end by our live AEGIS security platform. All IP in this programme belongs to JDB Sales, licensed to our UK holding company.

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### SLIDE 2

[CONTEXT — 2 min] Set the frame before numbers: the NHS already has the App (39m+ users), NHS Login, and the NHS Number — three of the four layers of a national identity. The missing fourth is a national credential: a mobile secure-enclave plus a low-cost physical token for those who can't use a phone. Estonia is the proof this works: 98% ID coverage, 99% of health processes digital, 1,345 years of citizen time saved per year. And we are ready now: DIATF alpha design exists, OfDIA and the ICO have been pre-notified, both certifying bodies are engaged, and AEGIS is live and tested.

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### SLIDE 3

[ESTONIA — 3 min] Give the comparator teeth. 98% card coverage, 99% health digital, 100% e-prescriptions, 1,345 years saved. Then the three economic lessons: the card is a low-cost public utility — value comes from the permissions economy at the edges (e-Residency returned ~EUR 125m in a single year); identity was built BEFORE health (the EHR rode the 2002 card); and X-Road means no central mega-database — the once-only principle is law. Same philosophy, tailored to the UK.

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### SLIDE 4

[FOUNDATION — 2 min] Walk the four layers. 1 NHS App — the digital front door. 2 NHS Login — the national identity service for health. 3 NHS Number — the universal identifier plus staff Care Identity Service. 4 — the missing piece we build: the National Credential on a secure rail. The fourth layer unlocks once-only, citizen audit rights, and a single trusted key to public services. This is a layer over what exists — nothing is replaced, nothing is ripped out.

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## **SLIDE 5**

[PARTNERSHIP — 3 min] This is the philosophy slide. We are a single accountable delivery partner; IP stays with JDB Sales, licensed to the holding company. Hybrid credential: physical smart card + wallet/app for ~50 million citizens. AEGIS is woven through — live, tested, DIATF/DSPT/DTAC-ready — which avoids a 3-5 year bespoke security build. Estonia is the proven comparator. And revenue is honest: ~£24.4bn ten-year value, ~£155-360m programme cost, ~28x ROI — with a permission-economy revenue stream at the edges (verification APIs). Certified UK/EU bureaus and HSMS handle physical manufacturing at volume.

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## **SLIDE 6**

[PROGRAMME 01 — 3 min] The core programme. 50 million smart cards + app; £24.4bn ten-year value; 28.2x projected ROI. Hybrid credential: mobile secure-enclave for the majority, low-cost physical token for the digitally excluded — nobody left behind. Two mandates worth emphasising: the citizen transparency mandate (every data access logged, one-touch red-flag to the ICO) and once-only by law (the state may not ask for information it already holds). Built to DIATF gamma 0.4 rules from day one.

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## **SLIDE 7**

[PROGRAMME 02 — AEGIS — 3 min] This is what nobody else in the room has. AEGIS is live today: 8 security modules, 37 endpoints, 24 defence layers, 8/8 audit findings fixed. It covers all 7 AI attack vectors — prompt injection, agent authorisation, MCP gateway, RAG, supply chain, model extraction, vector security. A 24/7 AI guardian that detects, blocks, reports — producing the audit evidence regulators want. GDPR/CE-ready, DSPT self-assessed, API-first, live in days. Independently tested live: 27/27 injection blocks, zero false positives; auth enforced 401/401/200.

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## **SLIDE 8**

[COMPLIANCE — 2 min] The path is real, not aspirational. DIATF gamma 0.4: alpha built to government rules; certification with BSI/Kantara in motion. OfDIA + ICO: pre-notification sent 27 Aug, DPIA in progress, ICO register reference ZC230615. Both CABs — BSI and Kantara — engaged. DSPT v8 / DTAC v2: AEGIS self-assessed, evidence pack in place. And the GOV.UK digital identity register entry is prepared for post-certification. If asked: we're doing this ourselves, we're not looking for the certification to be done for us.

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## SLIDE 9

[ROADMAP — 2 min] Four phases, every one gated on evidence. 2026: engage and plan — partnership, DIATF sandbox, RFI responses, certification kick-off. 2026-27: the pilot — regional issuance (card + app), AEGIS live, bureau contracts, CMS + HSM. 2027-29: scale — national rollout of 50M, wallet + attributes, cross-sector joins. 2029-30: the ecosystem — permissions economy, GOV.UK One Login, full coverage. No phase moves without certification, pilots, compliance, and live deployments.

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## SLIDE 10

[COMMERCIAL — 2 min] Honest economics. AEGIS is commercial today: Essentials £5,000 · Advanced £30,000 · Enterprise £100,000 per year — that funding keeps the corridor alive while the programme certifies and pilots. The licensing model: JDB Sales owns the IP, licences to the holding company; per-card, per-user, per-API transaction licensing at scale. And the NHS value: reduced duplicated data, freed clinical time, citizen empowerment, a consent-first and audit-visible health identity.

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## SLIDE 11

[THE ASK — 3 min] Close with four concrete asks. One: agree the partnership model — supplier-of-record or licensed delivery partner — with NHS England commercial. Two: sponsor a DIATF-compliant identity wallet plus AEGIS security pilot within Product & Platforms. Three: route through DHSC / NHS England commercial for a funded programme — the £24.4bn / 28x ROI case is on the table. Four: stand up a data & security working group — AEGIS plus NHS cyber — to run the DTAC/DSPT evidence. Ask which of the four they'd like to move first.

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## SLIDE 12

[CLOSE — 1 min] Thank them. One sentence summary: the identity-first future for the NHS, secured end to end by AEGIS — a partnership, a pilot, and a proven blueprint. Offer the leave-behind (one pager) and the NDA if they want the architecture detail. Give your contact: jdbsale3@gmail.com, 01922 445318. Invite next steps within two weeks.

**END — offer leave-behind + NDA; follow up within 2 weeks.**

# Atamis Registration Email — ZEUS AI Intelligence (NHS Health ID Wallet)

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## ATAMIS REGISTRATION EMAIL — NHS England eCommercial (draft)

**Why:** Atamis is the NHS England eCommercial supplier portal. The meeting deck references supplier registration; this email either (a) follows your Atamis account creation to confirm onboarding, or (b) is the engagement opener to NHS England Commercial once registered. Two variants below — use the one that matches your stage.

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### Variant A — sent AFTER you create the Atamis account (recommended)

**To:** england.supplier@nhs.net **Subject:** Supplier registration complete — ZEUS AI Intelligence (NHS Health ID Wallet, pilot-first)

> Dear NHS England Commercial Team, > > I am writing to confirm that ZEUS AI Intelligence (JDB Sales) has registered as a supplier on the NHS England eCommercial portal (Atamis) and would welcome engagement with your team. > > **Who we are** > ZEUS AI Intelligence is a UK company (ZEUSTRUSTAEGISSECURITY LTD, Company No. 17391549, 66 Paul Street, London EC2A 4NA). We develop a sovereign NHS health identity solution — a patient-owned ID wallet and physical card secured by our own AEGIS AI Security product — as a pilot-first proposition. > > **Why we are contacting you now** > - We are building our compliance position before any public promotion: DIATF alpha design (gamma 0.4), OfDIA/ICO pre-notification sent, DPIA in progress, both approved CABs (BSI and Kantara) engaged. > - We wish to align with the correct route to market: best-fit framework (SBS10521 / G-Cloud 14), the DSPT / DTAC / DARS pathway for a pilot trust or ICB, and DAPB3051 identity-verification standards. > - We would welcome a short conversation with the relevant commercial or digital-identity lead, and are ready to provide our compliance position and product details under appropriate arrangements (NDA available on request). > > An executive summary of the proposition is available on request, and our 1 October briefing deck summarises the partnership model. > > Kind regards, > > **Darren Birch** > Founder & Developer, ZEUS AI Intelligence > ZEUSTRUSTAEGISSECURITY LTD (Company No. 17391549) > jdbsale3@gmail.com · 01922 445318 · X: @jdbsales3

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### Variant B — pre-registration engagement (if you have not yet created the Atamis account)

**To:** england.supplier@nhs.net **Subject:** Pre-registration enquiry — ZEUS AI Intelligence (NHS Health ID Wallet, pilot-first)

> Dear NHS England Commercial Team, > > I am Darren Birch, founder of ZEUS AI Intelligence (JDB Sales), a UK company. We are developing an NHS health identity wallet — a patient-owned identity solution secured by our AEGIS AI Security product — as a strategic, pilot-first product. > > Before completing our supplier registration on Atamis, we would welcome guidance on: > 1. The correct supplier onboarding path and any commercial briefings available > 2. Best-fit framework and the letting/opportunity calendar (SBS10521 / G-Cloud 14) > 3. The DSPT / DTAC / DARS pathway for a pilot trust or ICB > 4. Alignment to DAPB3051 identity verification & authentication standards > 5. The SME Advisory Group and relevant contacts > > We are not yet publicly promoting the product; we are building our compliance position and would welcome a conversation with the relevant commercial team. > > Kind regards, > > **Darren Birch** > Founder & Developer, ZEUS AI Intelligence > ZEUSTRUSTAEGISSECURITY LTD (Company No. 17391549) > jdbsale3@gmail.com · 01922 445318

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## Attachments (send with Variant A)

- [ ] ZEUS-NHS-EXECUTIVE-SUMMARY-1PAGE.pdf
- [ ] NHS-Leave-Behind-ZEUSTA.pdf
- [ ] (Optionally) NHS-10ct-Call-Deck-ZEUSTA.pdf — or wait and send after the call

## Atamis portal reference

- Portal: <https://atamis-2463.my.site.com/s/welcome>
- Account: create supplier profile → complete registration → set tender alerts for "identity", "wallet", "DAPB3051"
- Keep the portal reference number from the confirmation email — quote it in Variant A if you have it

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All IP belongs to Darren Birch — ZEUSTRUSTAEGISSECURITY LTD (administered by the Darren & Jill Birch Trust).

# NHS ID Card System — Information Architecture

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## NHS ID CARD SYSTEM — INFORMATION ARCHITECTURE

### The Structural Blueprint of the UK National Digital Identity & Health-Card System

**Owner:** ZEUSTRUSTAEGISSECURITY LTD (Company No. 17391549) **Prepared by:** Darren Birch — ZEUSAI / JDB Sales **Date:** 27 August 2026 **Basis:** Estonia eID/e-Health evidence base · UK NHS digital landscape · AEGIS AI Security layer **Status:** Live working document — source of truth for the build

*All IP belongs to JDB Sales.*

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## 0. DOCUMENT MAP (How This Architecture Fits Together)

This information architecture is the master index for the NHS ID Card System. Every component of the programme is catalogued here — layers, modules, data objects, stakeholders, compliance frameworks, deployment options, and phasing — so that any document, spec, or build task can be located and traced back to the overall structure.

### NHS ID CARD SYSTEM

- 1. Programme Vision & Objectives
  - 2. Foundation Layers (the 4 layers)
  - 3. System Architecture (components & data flow)
  - 4. Identity Model & Card Specification
  - 5. Data Exchange Layer (UK X-Road)
  - 6. Health Integration
  - 7. Security & Compliance Framework
  - 8. Deployment Options
  - 9. Stakeholder Map
  - 10. Governance & Legal
  - 11. Programme Delivery Phases
  - 12. Metrics & Success Measures
  - 13. Reference Library (documents, CDNs, assets)
  - 14. Appendices
-

# 1. PROGRAMME VISION & OBJECTIVES

## 1.1 Vision statement

A single, trusted, national digital identity for every UK citizen — the key that unlocks health, benefits, tax, and public services once-only, securely, and transparently — secured by an AI-native security layer and proven by the Estonian model.

## 1.2 Core objectives

| #  | Objective                   | Measure                                                     |
|----|-----------------------------|-------------------------------------------------------------|
| O1 | Universal identity coverage | ≥90% of adult population enrolled within 5 years            |
| O2 | Once-only data collection   | ≤1 repeated data request per citizen per transaction set    |
| O3 | Digital health access       | 100% of prescriptions digital; full EHR access for citizens |
| O4 | Transparency                | 100% of data-access events logged and citizen-visible       |
| O5 | Security at scale           | Zero preventable major breaches; 24/7 AI-native monitoring  |
| O6 | Inclusion                   | Physical fallback for every digitally excluded citizen      |

## 1.3 The four foundational layers (already present in the NHS — the missing layer is ours)

**NHS App** — ~39M users, digital front door (exists)

**NHS Login** — national identity service for health (exists)

**NHS Number + staff CIS** — universal identifier (exists)

**National digital identity + physical/mobile credential** — THE LAYER WE ADD (gap)

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# 2. FOUNDATION LAYERS

LAYER 4    IDENTITY & CREDENTIAL    ■■■ NEW — the NHS ID Card / mobile passkey / physical token  
LAYER 3    DATA EXCHANGE (UK X-Road)    ■■■ NEW — open, decentralised transport  
LAYER 2    HEALTH & PUBLIC SERVICES    (existing: NHS App, Spine, NHS Login)  
LAYER 1    LEGAL & TRUST    (Digital Identity Act, DIATF, DPA 2018, NIS)

Each layer has owners, standards, and dependencies:

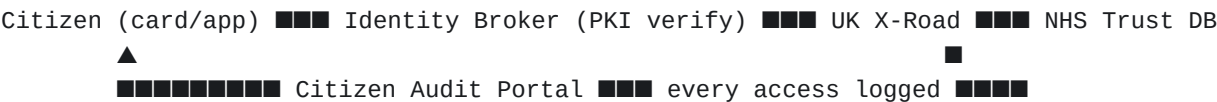
- **Layer 4 (identity)** depends on Layer 1 (legal validity) and Layer 3 (exchange).
- **Layer 3 (exchange)** depends on Layer 1 (NIS/DIATF compliance).
- **Layer 2** is largely existing but must be re-pointed at Layers 3–4.

### 3. SYSTEM ARCHITECTURE — COMPONENTS & DATA FLOW

#### 3.1 Components

| Component             | Type           | Description                                                             | Status   |
|-----------------------|----------------|-------------------------------------------------------------------------|----------|
| NHS ID Credential     | Card/app/token | Citizen identity token (dual-key chip or secure-enclave)                | NEW      |
| UK X-Road             | Data layer     | Open, decentralised, encrypted data exchange (MIT-licensed X-Road core) | NEW      |
| Identity Broker       | Service        | Routes authentication; cannot see clinical payloads (tokenisation)      | NEW      |
| Citizen Audit Portal  | Service        | Every access event logged & visible to citizen; right to challenge      | NEW      |
| NHS Spine integration | Existing       | NHS national data spine — connects via the new layer                    | EXISTING |
| NHS App               | Existing       | Digital front door — becomes the mobile identity surface                | EXISTING |
| AEGIS AI Security     | Security       | 24/7 AI-native guardian over the whole stack (24 layers, 37 endpoints)  | PROVIDED |

#### 3.2 Data flow (authentication + health access)



- Identity broker authenticates; **cannot decrypt** clinical payloads (end-to-end tokenisation).
- Trusts keep their own data; exchange is authenticated, logged, once-only.
- AEGIS monitors every node, blocks AI-attacks in real time, and feeds the audit trail.

## 4. IDENTITY MODEL & CARD SPECIFICATION

### 4.1 *The credential (Estonia-informed, UK-evolved)*

- **Form factors:** (a) physical chip card (Estonian model — full sovereignty, offline photo-ID); (b) software secure-enclave on phone (Apple Secure Enclave / Android StrongBox) provisioned via passport-chip NFC; (c) low-cost hardware token (YubiKey-class) for the unbanked/digitally excluded.
- **Keys:** dual 2048-bit RSA (or ECC) key pairs — Key 1 authentication, Key 2 digital signature (legally equivalent to handwritten per the Digital Signatures Act model).
- **Hybrid strategy:** mobile-first for the majority → physical token for the excluded → full smart-card per national-security/offline requirements where needed.

### 4.2 *Enrolment & verification*

- NHS Login / passport biometrics / verified birth records as proofing sources (DIATF-aligned).
  - Assisted enrolment at GP surgeries, libraries, council service points.
- 

## 5. DATA EXCHANGE LAYER (UK X-ROAD)

### 5.1 *Principles*

- Decentralised: no central mega-database; each organisation owns its data.
- Open: X-Road core is MIT-licensed; adopt + extend.
- Once-Only: legally enforced — the state may not ask twice.
- Peer-to-peer mesh: one node down does not bring down the state.

### 5.2 *Standards*

- Uniform API data-exchange standard across councils, DWP, HMRC, NHS.
  - End-to-end tokenisation so the broker never sees the payload.
  - Interoperability with FHIR UK Core for health.
- 

## 6. HEALTH INTEGRATION

### 6.1 *Estonia-proven sequencing (identity first, health second)*

| Milestone             | Estonia | UK plan        |
|-----------------------|---------|----------------|
| Identity live         | 2002    | Phase 1        |
| EHR live              | 2008    | Phase 2–3      |
| 100% e-prescriptions  | 2010s   | Phase 3 target |
| Cross-border e-health | 2024    | Future-state   |

## 6.2 NHS use cases

- Repeat prescriptions without appointments
- Patient access to records, labs, letters (via NHS App + audit portal)
- Staff ID with role-based access (doctors' own credentials)
- Allied use: 2FA for staff, secure messaging, eConsent

# 7. SECURITY & COMPLIANCE FRAMEWORK

## 7.1 Threat model (AI-era)

Prompt injection, agent authorisation, MCP gateway, RAG security, supply-chain compromise, model extraction, vector security, plus classic network/API/access/data attacks. AEGIS covers all 7 AI attack vectors + 24 defence layers + 37 endpoints.

## 7.2 Regulatory landscape (beyond UK GDPR)

| Framework                                 | Requirement                           | Architecture response                                                                 |
|-------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------|
| DPA 2018 Sch 1 Pt 1                       | Strict conditions for health/SC data  | Public-health statutory basis; end-to-end tokenisation; decrypt only at point of care |
| UK NIS Regulations                        | Critical infrastructure security      | Peer-to-peer mesh; no single point of failure; AEGIS monitoring at every node         |
| DIATF (DSIT)                              | Formal identity/attribute trust rules | Design to DIATF from day one; use the DIATF sandbox for alpha                         |
| CLDC (Common Law Duty of Confidentiality) | Patient confidentiality               | Granular consent dashboard; revocable permissions; binding audit trail                |

## 7.3 AEGIS role

24/7 AI guardian; audit-evidence builder; GDPR/CE-ready, DSPT self-assessed, 8/8 findings fixed; live demo + health check (see Reference Library).

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## 8. DEPLOYMENT OPTIONS

### 8.1 Option A — *Physical smart-card (pure Estonian model)*

- Full sovereignty, offline photo-ID, high hardware security
- Massive rollout cost, political resistance (Identity Cards Act 2006 repeal precedent)

### 8.2 Option B — *Software secure-enclave (UK evolution)* ← **default recommendation**

- Near-zero distribution, instant reach to 85%+ via existing phones, consumer-native
- Digital exclusion; dependence on Apple/Google lifecycles → mitigated by hybrid token

### 8.3 Recommended: *Hybrid*

Mobile-first + low-cost hardware token for the excluded + card/physical where required.

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## 9. STAKEHOLDER MAP

| Stakeholder                        | Interest                      | Engagement                          |
|------------------------------------|-------------------------------|-------------------------------------|
| NHS England (digital + commercial) | Service delivery, procurement | Primary commercial route            |
| DHSC                               | Policy, mandate               | Ministerial/sponsor briefings       |
| DWP, HMRC                          | Cross-service identity        | Once-only joins                     |
| DSIT                               | DIATF certification           | Standards engagement                |
| ICO                                | Data protection regime        | Pre-notification consultation       |
| CQC / care sector bodies           | DSPT, standards               | AEGIS care-sector channel           |
| Providers & citizens               | Usability, trust, inclusion   | Public dialogue, assisted enrolment |
| Parliament                         | Legislation                   | Digital Identity Act sponsorship    |

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## 10. GOVERNANCE & LEGAL

### 10.1 Instruments needed

**Digital Identity & Cryptographic Signatures Act** — grants e-signatures legal validity; establishes once-only; mandates audit rights.

**DIATF certification** for the identity service.

**NIS-registered operator** status for the exchange layer.

**Data protection impact assessment (DPIA)** — DPA 2018 compliant.

**Trust framework + Caldicott** governance for health data.

### 10.2 Programme governance

- Joint NHS England / DHSC / DSIT board; ZEUS (“JDB Sales” licensing, holding co ZEUSTRUSTAEGISSECURITY LTD) as delivery partner; AEGIS as security layer; independent assurance; citizen panel.
- 

## 11. PROGRAMME DELIVERY PHASES (see separate UK Adoption Roadmap doc)

| Phase | Focus                        | Horizon      |
|-------|------------------------------|--------------|
| 0     | Mobilise & sandbox (DIATF)   | 0–3 months   |
| 1     | Legal & identity layer       | 3–12 months  |
| 2     | UK X-Road pilot (single ICS) | 6–18 months  |
| 3     | Health integration & scale   | 12–36 months |
| 4     | Cross-sector & national      | 36–60 months |

*Full detail: “UK Adoption Roadmap” document (same release).*

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## 12. METRICS & SUCCESS MEASURES

- Adoption: % enrolled, % active monthly, % using health services
- Once-only: duplicated-request count per citizen-year
- Time saved: modelled citizen-time saved per year (Estonia: 1,345 yrs/yr)
- Trust: audit-challenge rate, opt-out rate, satisfaction
- Security: time-to-detect, breaches prevented, audit completeness

- Economics: cost per enrolment, ecosystem revenue (mirrors €125m e-Residency model)
- 

## 13. REFERENCE LIBRARY

### 13.1 Core documents (*this release*)

| Doc                           | File/CDN                                           |
|-------------------------------|----------------------------------------------------|
| This information architecture | (master doc)                                       |
| 20-page research paper        | CDN aa0f41b7                                       |
| Executive summary             | CDN (this release)                                 |
| UK adoption roadmap           | CDN (this release)                                 |
| Estonia one-pager             | /home/user/zeus-assistant-v2/estonia-one-pager.txt |
| Estonia standalone case PDF   | CDN 8c552fec                                       |
| AEGIS enterprise letter       | CDN 3faca24d (PDF)                                 |
| NHS briefing pack             | zeusaiintelligence.com/briefing.html               |

### 13.2 Live assets (*verified*)

- AEGIS demo: <https://aegis-security.higgsfield.app>
- AEGIS health: <https://apiaegissecurity.tech/health>
- NHS thread ([lauren.roberts3@nhs.net](mailto:lauren.roberts3@nhs.net), 25–26 Aug) — briefing pack shared

### 13.3 Key figures (*as-cited, directional*)

98% ID coverage · 99% digital health processes · 1,345 yrs/yr saved (Estonia) · €125m/yr e-Residency revenue · NHS App ~39M users

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## 14. APPENDICES

- A. Estonia timeline (1991–2024)
  - B. Glossary (eID, X-Road, PKI, DIATF, CLDC, DSPT, ICS)
  - C. Regulatory compliance checklist
  - D. Risks & mitigations register (start)
- 

**END OF INFORMATION ARCHITECTURE — All IP belongs to JDB Sales.**

## NHS ID CARD + AEGIS — DETAILED PROJECT PLAN

Project: National NHS Digital Identity & Card (50M citizens) · Owner: ZEUSAI / ZEUSTRUSTAEGISSECURITY LTD Date: 28 Aug 2026 · Basis: capability breakdown + partnership proposal · All IP belongs to JDB Sales.

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### 1. OBJECTIVES & SUCCESS CRITERIA

- Issue a hybrid NHS credential (physical chip card + mobile wallet/app) to 50M citizens.
- Identity-first, Estonia-proven; AEGIS-secured end to end.
- Fully DIATF / GPG45 / DSPT / DTAC compliant; UK data residency; ICO registered.
- Deliver ~GBP 24.4bn/10-yr value at ~28x ROI; permissions-economy revenue model.

Success gates: (1) NHS partnership + commercial route signed; (2) DIATF certification (IDSP/HSP/ASP) via BSI/Kantara; (3) pilot issued; (4) national rollout milestone; (5) DSPT/DTAC evidence accepted.

---

### 2. WORKSTREAMS (WBS high-level)

| WS | Workstream                                                       | Owner           | Depends on |
|----|------------------------------------------------------------------|-----------------|------------|
| 0  | Programme governance & NHS partnership                           | Darren          | —          |
| 1  | Compliance & certifications (DIATF/GP G45/DSPT/DTAC/ICO)         | Compliance      | WS0        |
| 2  | Identity & security architecture (AEGIS, PKI, HSMs, DIATF alpha) | Engineering     | WS1        |
| 3  | Digital identity wallet / app + NHS integration                  | App team        | WS2        |
| 4  | Physical card: bureau procurement + CMS + HSM                    | Procurement/Eng | WS1, WS3   |

| WS | Workstream                         | Owner      | Depends on |
|----|------------------------------------|------------|------------|
| 5  | Enrolment & fulfilment operations  | Ops        | WS4        |
| 6  | Data, fraud & risk                 | Fraud/Risk | WS2        |
| 7  | Pilot & rollout, training, go-live | PMO        | all        |

### 3. PHASES, MILESTONES & TIMELINE

#### ***Phase 1 — Engage & Plan (Month 0–6)***

Tasks:

- 1-Oct NHS call → partnership agreement + commercial route (M1)
- DIATF sandbox build (continue) + OfDIA/ICO engagement (M1–3)
- RFIs to bureaus (issued M1) → shortlist + NDA + per-card quotes (M2–3)
- DIATF certification kick-off with BSI/Kantara (M1–3)
- DPIA, RoPA, ICO, UK-residency sign-off (M3–6)
- Prove AEGIS + DIATF alpha (M3)

Milestone M1: Partnership agreed + commercial route + DIATF sandbox live.

#### ***Phase 2 — Pilot (Month 6–18)***

- Contract bureau (Certus/Entrust/Thales/G+D) for pilot (M6–8)
- Provision HSMS + key ceremony; stand up CMS + PKI (M6–9)
- Build wallet MVP + NHS App/Spine/FHIR integration (M6–12)
- Pilot issuance: 1k–10k cards + app in one region (M9–14)
- Fraud/Risk + consent/audit-trace live (M12)
- AEGIS full coverage on pilot (M12)

Milestone M2: Pilot issued + DIATF certification + DSPT/DTAC evidence drafted.

#### ***Phase 3 — Scale (Month 18–36)***

- National bureau capacity + fulfilment (M18)
- Regional rollout wave-by-wave (M18–30)
- Mobile wallet + attributes (vaccination, e-prescription) (M20–30)
- Cross-sector (DWP/HMRC/banking) joins + permissions APIs (M24–36)
- SIEM/NIS/SOC, pen-test, IR (M24–30)
- DTAC v2 / DSPT v8 acceptance (M30)

Milestone M3: 50% national coverage + ecosystem API live.

#### ***Phase 4 — National / Ecosystem (Month 36–60)***

- Full 50M issuance + wallet
- Permissions-economy + GOV.UK One Login alignment
- Renewals, replacements, inclusive enrolment (offline/assisted)
- NIS/operational excellence

Milestone M4: National completion + value case delivered.

---

## **4. KEY TASKS / DEPENDENCIES (critical path)**

NHS partnership (M1) → commercial route → contract

DIATF certification → enables live identity service

Bureau RFI → contract → card personalisation capacity

HSM + key ceremony → card signing

Wallet + NHS integration → citizen access

Enrolment/fulfilment ops → issue

Dependency rule: compliance gate (DIATF, DSPT, DTAC, DPIA) then security (AEGIS) then issue.  
No issuance without the gate+security both green.

---

## **5. REGISTER / RISK (top 6)**

| <b>Risk</b>                               | <b>Impact</b> | <b>Mitigation</b>                                                     |
|-------------------------------------------|---------------|-----------------------------------------------------------------------|
| Political/public identity-card resistance | High          | transparency mandate, decentralised, voluntary framing, citizen panel |
| DIATF certification delay                 | High          | engage CABs early, sandbox evidence from M1                           |
| Bureau capacity/pricing at 50M            | Med           | multi-bureau competition, RFI now, staggered scale                    |
| HSM/key-ceremony security failure         | Critical      | M-of-N, audited ceremony, dual HSM                                    |
| NHS procurement timeline                  | Med           | SBS/DHSC commercial route in parallel                                 |
| Data residency/fraud                      | Med           | UK-cloud, CIFAS/LexisNexis, AEGIS fraud layer                         |

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## 6. GOVERNANCE & CADENCE

- Steering: monthly (Darren + NHS/DHSC sponsor)
  - Workstream leads: weekly
  - Compliance board: monthly
  - Escalation: any security/certification issue to programme lead within 24h
- 

## 7. ESTIMATED TOP-LEVEL COST (see costing in the full check)

- In-house platform+app+PKI: £10–20M
- HSMS + secure infra: £0.5–2M
- DIATF/audits/certs: £0.5–1.5M
- Card+fulfilment (contract, scaled): per-card cost × volume (dominant)
- AEGIS ops: ongoing
- Total programme (50M): ~£250M–£600M+ (per capability breakdown)

END — All IP belongs to JDB Sales.

## NHS PARTNERSHIP PROPOSAL

### ZEUSAI / ZEUSTRUSTAEGISSECURITY LTD — National Digital Identity & NHS ID Card System

Prepared: 28 Aug 2026 · For: NHS England (Product & Platforms: Stephen Koch, Helen Clifton, Paul Wilkinson) · Meeting: Thu 1 Oct 16:00–16:30 Company: ZEUSTRUSTAEGISSECURITY LTD (No. 17391549) · ZEUSAI / JDB Sales · ICO ref ZC230615 All IP belongs to JDB Sales.

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#### 1. ONE-LINE POSITION

A partnership to deliver the national NHS digital identity (50M citizens) — a hybrid physical smart-card + NHS app/wallet — built on the proven Estonia model, secured end-to-end by AEGIS AI Security, fully DIATF/GPG45/DSPT/DTAC-compliant, from planning → negotiation → pilot → national.

#### 2. WHAT WE PROPOSE (the partnership)

| Element        | Detail                                                                                                                                                                       |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Deliverable    | NHS ID Card (physical chip card) + NHS digital identity wallet/app for ~50M citizens, with attributes (health, vaccination, e-prescription) and citizen transparency mandate |
| Security layer | AEGIS AI Security — live, tested: all 7 AI attack vectors, 24 defence layers, 37 endpoints; DSPT/DTAC-ready                                                                  |
| Identity model | Estonia-proven: 98% coverage, 99% digital, once-only, decentralised X-Road-style, citizen audit trail                                                                        |
| Value case     | ~GBP 24.4bn / 10-yr value, ~28x ROI; Estonia e-Health comparator (EUR 125m/yr e-Residency model)                                                                             |
| Compliance     | DIATF (legally recognised under UK Data Act), GPG45/44, DSPT v8, DTAC v2, ICO registered, UK data residency                                                                  |

| Element        | Detail                                                                                                                                                   |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Delivery model | ZEUSAI owns IP, programme, wallet, AEGIS, NHS integration, compliance; certified UK/EU bureaus + HSMs handle physical manufacturing/fulfilment at volume |

### 3. THE FOUR LAYERS (where we add the missing 4th)

NHS App (39m+ users) — exists

NHS Login — exists

NHS Number + staff CIS — exists

National identity credential (physical + mobile) + secure rail — WE SUPPLY (AEGIS-secured)

### 4. WHY NHS ENGLAND SHOULD PARTNER (not procure separately)

- Speed: we bring a live, tested platform + DIATF path + a ready security layer (no 3–5 yr build/cert of a bespoke security stack).
- IP + accountability: single accountable partner (holding co No. 17391549) owning end-to-end delivery and security.
- Proven comparator: identity-first playbook validated by Estonia (and request "the Estonia case" they asked about).
- Revenue: permissions-economy model — identity as a low-cost public utility, monetised at the edges (verification APIs), mirroring Estonia's €125m/yr.
- Security by default: AEGIS is already DTAC/DSPT-aligned and independently tested — de-risks the cyber posture for a 50M-scale identity programme.

### 5. PHASED ROADMAP

| Phase             | Horizon  | Outcome                                                                          |
|-------------------|----------|----------------------------------------------------------------------------------|
| 1 · Engage & plan | 0–6 mo   | Partnership agreement, DIATF sandbox, proof-of-concept with NHS, ICO/DPIA        |
| 2 · Pilot         | 6–18 mo  | Regional issuance (physical + app), AEGIS live, bureau contracts, CMS + HSM live |
| 3 · Scale         | 12–36 mo | National rollout, 50M cards + wallet, DSPT/DTAC evidence, cross-sector joins     |

| Phase                    | Horizon  | Outcome                                                                           |
|--------------------------|----------|-----------------------------------------------------------------------------------|
| 4 · National / ecosystem | 36–60 mo | Full coverage, permissions economy, GOV.UK One Login alignment, NIS/SOC readiness |

## 6. THE ASK (1 Oct)

Confirm partnership model (supplier-of-record / licensed-delivery-partner) with NHS England commercial.

Sponsor a DIATF-compliant identity wallet + security pilot within Product & Platforms.

Route through DHSC / NHS England commercial for a funded programme (see value case).

Agree a data/security working group (AEGIS + NHS cyber) to run DTAC/DSPT evidence.

## 7. OUR COMMITMENTS

- All IP belongs to JDB Sales; licensed to the holding company; programme delivered under NHS governance.
- Security (AEGIS) is independently verifiable; we'll run a live demo in the first session.
- Full DIATF, DSPT, DTAC, ICO, UK-residency compliance in the plan.

NEXT STEP: 30-min call 1 Oct → agree partnership model + pilot scope + commercial route.

END — All IP belongs to JDB Sales.

# AEGIS AI Security — NHS Installation & Integration Runbook

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## AEGIS AI SECURITY — NHS INSTALLATION & INTEGRATION RUNBOOK

**Step-by-step: install AEGIS into NHS services and integrate it with the NHS ID Card System**

**Company:** ZEUSTRUSTAEGISSECURITY LTD (17391549) · ZEUS AI / JDB Sales **Product:** AEGIS AI Security — 8 modules · 37 endpoints · 24 defence layers · 7 AI attack vectors **Audience:** NHS Trust / ICS IT & security leads, commissioning/commercial teams **Version:** NHS-I v1.0 · 27 August 2026 · All IP belongs to JDB Sales

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## PART A — READY: PRE-INSTALLATION (the 4 NHS gates)

Before any server work, complete these four gates. AEGIS is "ready to install" only when all four are green.

### Step 0 — Governance & compliance gates

| Gate     | Action                                                                                                             | Owner            | Evidence                            |
|----------|--------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------|
| 1 · DPIA | Submit/complete the Data Protection Impact Assessment for the security service (health/SC data, DPA 2018 Sch1 Pt1) | Trust DPO + ZEUS | DPIA record, ICO pre-notification   |
| 2 · DSPT | Confirm AEGIS self-assessment aligns to your DSPT submission                                                       | Trust IG         | DSPT tool entries referencing AEGIS |

| Gate                                    | Action                                                                                            | Owner          | Evidence                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------|----------------|------------------------------|
| 3 · NIS                                 | Register/confirm AEGIS under relevant NIS framework as critical-infrastructure security component | Trust/ICS      | NIS register + data-flow map |
| 4 · DIATF (for NHS ID Card integration) | Keep AEGIS aligned to the DIATF alpha of the identity programme (OfDIA/ICO pre-notified)          | Programme lead | DIATF design doc             |

**Contract & commercial:** route via G-Cloud 14 (Lot 4 Cloud Security) or CCS/THS framework; pricing Essentials £5,000 / Advanced £30,000 / Enterprise Managed Guard £100,000 pa via JDB Sales; DSAR/data-controller line: ZEUSTRUSTAEGISSECURITY LTD (ICO C2015509).

## Step 1 — Choose the deployment model

| Model                                    | Suits                                   | Notes                                                                                                            |
|------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>SaaS (hosted)</b>                     | Pilots, quick proof-of-concept          | Fastest; uses <a href="https://apiaegissecurity.tech">https://apiaegissecurity.tech</a> ; free key = 1000 req/mo |
| <b>Self-hosted (recommended for NHS)</b> | Trust environments                      | Run AEGIS <b>on your own infrastructure</b> ; data stays in-house; full control                                  |
| <b>On-premise/air-gapped</b>             | Highly sensitive / disconnected estates | No external calls; local models/keys                                                                             |

**Recommendation for NHS:** Self-hosted on Trust/ICS infrastructure (or NHS-approved cloud region), to keep patient data routing fully in-house and satisfy DSPT/NIS residency expectations.

## Step 2 — Confirm infrastructure (self-hosted)

- Ubuntu 24.04 LTS (or your standard NHS Linux build) · **2 vCPU / 4 GB RAM / 40 GB** min
  - Ports **22 / 80 / 443** open (SSH admin + HTTPS service)
  - DNS: e.g. `aegis.[trust].nhs.uk` (or internal CNAME) — certificate via internal CA or Let's Encrypt on public/edge
  - PostgreSQL + Redis available (or installed by the installer), NGINX as reverse proxy
  - Asset location agreed: which team owns the AEGIS host, patching, and response (on-call) rota
-

# PART B — INSTALL (the 6-step procedure, from the AEGIS quick-start + operations manual)

## Step 3 — Prepare the host

```
# run as root on Ubuntu 24.04
hostnamectl set-hostname aegis-server
apt-get update && apt-get upgrade -y
# install the AEGIS stack: python venv/pip, reverse proxy, db, cache, tls, git
apt-get install -y python3-venv python3-pip nginx postgresql redis-server certbot git curl
```

> Use the Trust's standard image/patching baseline where it differs; keep these packages in the local mirror.

## Step 4 — Deploy the AEGIS release

```
mkdir -p /opt/aegis/backend /opt/aegis/data
# (Copy the AEGIS release provided by ZEUS/JDB Sales into /opt/aegis/backend/)
cd /opt/aegis/backend
python3 -m venv venv
./venv/bin/pip install -r requirements.txt
# configure .env with your DB + Redis + secret keys + allowed origins
```

Backend is a **FastAPI (Python)** service aggregating the 8 modules / 37 endpoints.

## Step 5 — Configure services & TLS

```
# enable + start database and cache
systemctl enable --now postgresql redis-server
# create the AEGIS systemd service (backend) and enable it
# - systemd unit runs the FastAPI app; expose on 127.0.0.1:8000
# reverse proxy via nginx from :443 -> :8000
# TLS certificate:
#   - internal: install your NHS internal CA cert
#   - edge/public: certbot --nginx -d aegis.[your-domain]
```

> **Residency rule:** keep the certificate, keys and all traffic inside your trust boundary. No patient data leaves the environment.

## Step 6 — Run the backend & verify the API

```
systemctl enable --now aegis
curl -sk https://aegis.[your-domain]/health
# expect: {"status":"healthy","service":"AEGIS","version":"1.0.0"}
```

Issue your first API key (registration endpoint or admin), then smoke-test in the next step.

## Step 7 — Integration of the 7 attack vectors (per service)

Wire AEGIS into each AI surface you run:

**Prompt defence** → intercept outgoing user prompts to LLMs (block @ score threshold)

**Agent authorisation / IAM** → policy filter before agents call tools

**MCP gateway** → route tool connections through AEGIS verification

**RAG security** → vet retrieval + grounding data before use

**Supply chain** → scan packages/images on ingest (CVE findings auto-flag)

**Model extraction** → monitor inference patterns for theft signals

**Vector security** → encrypt + DP on embeddings/vector store access

**Engineers:** add AEGIS SDK/agent (Node/Python/Java/Go) or call the API with `x-api-key`. Every request returns a verifiable `request_hash` — your audit-trail feed.

## Step 8 — Integrate with the NHS ID Card System (the flagship)

**Identity broker protection** — place AEGIS in front of the NHS identity broker; end-to-end tokenisation means AEGIS sees metadata/verdicts, never clinical payloads.

**Audit feed into the transparency mandate** — forward AEGIS request hashes + verdict logs to the citizen audit portal (every access logged, citizen-visible, red-flag to ICO).

**Before national rollout** — the certificate + DPIA + pen-test + production SLA gates (Part A) must all be green.

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# PART C — OPERATE (go-live readiness)

## Step 9 — Security hardening & monitoring

- Enable **AEGIS self-protection** ("AEGIS on itself") on its own host
- Forward AEGIS logs to your SIEM; set incident email / 24-7 managed-guard notifications (Enterprise tier)
- Regular: patch host (same baseline as Trust), rotate API keys, review triggered-rules thresholds
- 24/7 AI guardian already detects → blocks → reports; confirm alert routing to your on-call rota

## Step 10 — Verify & certify

| Check                              | Expected                                  |
|------------------------------------|-------------------------------------------|
| GET /health                        | healthy                                   |
| malicious prompt                   | verdict malicious → action block          |
| benign prompt                      | verdict safe → allow (no false positives) |
| supply-chain scan (vulnerable pkg) | findings + CVE + recommendation           |
| bad API key                        | HTTP 401                                  |
| Audit hashes                       | stable, queryable for ICO/DSPT evidence   |

## Step 11 — Go-live & support

- Publish the service to the Trust's service catalogue; DOCUMENT control: ZEUSTRUSTAEGISSECURITY LTD
  - Onboard teams (training pack available); agree SLA & support tier (Essentials/Advanced/Enterprise)
  - NIS/DSPT annual re-checks incl. AEGIS evidence; DIATF certification kicked off with BSI/Kantara
- 

## GATE CHECKLIST (printable sign-off)

- ☐ DPIA filed · ICO pre-notified
- ☐ DSPT self-assessment aligned
- ☐ NIS register entry / data-flow mapped
- ☐ Deployment model chosen (self-hosted recommended)
- ☐ Test key works (health + block + safe + CVE + 401)
- ☐ TLS + internal data residency confirmed
- ☐ Audit feed wired to transparency mandate
- ☐ Training delivered · SLA agreed · on-call set
- ☐ DIATF / pen-test / production SLA green for NHS ID Card rollout

**END OF RUNBOOK — All IP belongs to JDB Sales.**

# ZEUSAI — NHS ID Card + AEGIS

## Full Capability & Resource Breakdown

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### NHS ID CARD + AEGIS — FULL CAPABILITY & RESOURCE BREAKDOWN

**What ZEUSAI / ZEUSTRUSTAEGISSECURITY must have to run the project at 50M-citizen scale**

**Prepared:** 28 Aug 2026 · **Company:** ZEUSTRUSTAEGISSECURITY LTD (No. 17391549) · ZEUSAI / JDB Sales **Assumption set:** NHS partners the company to issue a national identity credential to **50M citizens** — a hybrid model (physical chip card **and** a mobile app/Digital Identity Wallet), secured end-to-end by AEGIS, compliant with UK DIATF/GPG45. All IP belongs to JDB Sales.

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#### 1. THE OPERATING MODEL (what "fully capable" means)

To issue ~50M cards you need **three integrated plants**: (a) secure physical-card production + personalisation, (b) digital identity/app + PKI issuance at scale, and (c) the security + compliance + operations fabric. You do **not** build all of it in-house; you hold the programme/IP/licensing and **contract the specialist industrial providers**, giving yourself the capability to run, verify and own the whole. That keeps IP with JDB Sales while the heavy manufacturing is done by certified bureaux.

Decision points this doc answers:

- **Physical cards via a card bureau** (UK/EU, certified secure) — not your own factory (capital-intensive, ~£millions, years to certify).
  - **Your in-house core:** program management, key management (HSMs + ceremony), the mobile app/identity wallet (you own the app + backend), AEGIS security layer, CRM/fulfilment control, compliance (DIATF/GPG45/DSPT/DTAC), and the NHS integration.
- 

#### 2. HARDWARE & EQUIPMENT (your in-house footprint)

*2a. Identity/card-side (mostly via bureaux, but you need reference capability)*

| Equipment                                                                 | Purpose                                                                                                      | Source                                  | Qty/notes                         |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------|
| Card printer (retail) – e.g. <b>Zebra</b> (ZXP/ZC series), <b>Entrust</b> | On-the-spot/pilot card issuance, small batch, lost-card replacement                                          | Zebra, Entrust                          | 2–4 units (pilots, local issuing) |
| Card lamination / overlay                                                 | Durability, security laminate                                                                                | Zebra / bureau                          | small volume only                 |
| Contact/contactless card reader + encoder                                 | Test/QA cards, in-house verification                                                                         | Gemalto/Thales, HID                     | 2–4                               |
| Chip/secure element for the card                                          | <b>Infineon SECORA ID</b> (CC EAL6+, Java Card 3.1), <b>NXP</b> SmartMX, optionally <b>ACOS</b> /Gemalto CEs | Infineon, NXP                           | contracted per batch              |
| Card body / ID1 PVC blanks + laminate                                     | Material supply                                                                                              | Certus, bureau partners, Zebra supplies | via bureau                        |
| Barcode/RFID/NFC readers (enrolment + verification)                       | Enrolment capture, field verification                                                                        | HID, IDemia, Zebra                      | regional estate                   |

**Reality:** printing 50M cards is a bureau job. Your in-house printers are for **pilots, instant issuance, replacements and QA**, not bulk.

## ***2b. Server / infrastructure (in-house or NHS-approved cloud)***

| Resource                         | Spec                                                                   | Note                                                                                          |
|----------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| HSMs (Hardware Security Modules) | <b>Thales Luna/Data Protection on Demand, Entrust nShield, Utimaco</b> | Host root CA + card signing keys; M-of-N key ceremony; required for WebTrust/ETSI/DIATF trust |
| HSM appliances                   | ≥2 (active + standby), plus cloud HSM                                  | Never expose keys in software                                                                 |
| Secure server estate             | NHS-approved/UK cloud (Azure UK South, AWS eu-west-2, or on-prem)      | Data residency UK                                                                             |
| PKI + certificate infrastructure | Issuing CAs, OCSP, CRL, key ceremony script                            | in-house                                                                                      |
| Card Management System (CMS)     | <b>Entrust Instant/central issuance, G+D, EastNets, Gemalto/Thales</b> | manages card lifecycle, personalisation jobs, activation, orders                              |
| Database + app backend           | PostgreSQL/SQL Azure, Redis                                            | scale out                                                                                     |
| Monitoring/security (AEGIS)      | already built                                                          | 24-7 AI-native layer                                                                          |

## 2c. Physical security / facilities

- Secure room / vault for HSM + media + blank cards (physical access control, CCTV, 2-person rule).
  - Data centre or NHS-approved cloud region (UK/EU data residency).
  - Mail/fulfilment handling areas if doing mail insertion in-house (else bureau).
- 

## 3. SOFTWARE STACK REQUIREMENTS

| Layer                         | What                                                                        | Vendors / open source                                                                                                                               | To build/buy                                                                    |
|-------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Digital Identity Wallet (app) | citizen app: passkey/credentials, NHS card + attributes, consent, audit log | <b>You own this</b> (Zeus app) → integrate; reference certified wallets: <b>OneID</b> (HSP/IDSP certified), <b>Post Office EasyID</b> , <b>Yoti</b> | <b>Build in-house</b> (your IP) — integrate certified providers for DIATF scope |
| Card Management System        | card lifecycle, personalisation, activation, ordering                       | Entrust, G+D, EastNets, Thales/Gemalto                                                                                                              | Buy / partner                                                                   |
| PKI / key management          | Root CA, issuing CA, key ceremony, HSM ops                                  | Thales, Entrust (nShield), Utimaco                                                                                                                  | Buy (HSM) + own processes                                                       |
| Identity proofing (IDSP)      | GPG45 identity checks, biometric liveness, document validation              | OneID, Yoti, Post Office, or build on GPG45                                                                                                         | Partner (DIATF cert)                                                            |
| Mobile provisioning           | passkey/Wallet pass issuance to app                                         | Apple/Google Wallet pass, or app-native                                                                                                             | Build/integrate                                                                 |
| CRM / case management         | enrolments, status, disputes, helpdesk                                      | Salesforce, HubSpot, ServiceNow, Zoho                                                                                                               | Buy                                                                             |
| Fulfilment / mailer           | address validation, print mail, batch mail                                  | Certus (fulfilment), Royal Mail / Whistl / DocPost                                                                                                  | Partner                                                                         |
| Fraud / risk                  | identity fraud checks, synthetic identity, sanctions                        | CIFAS (UK), LexisNexis, Experian, Onfido                                                                                                            | Partner/integrate                                                               |
| Security (AEGIS)              | all layers                                                                  | <b>AEGIS (yours)</b>                                                                                                                                | already built                                                                   |
| Observability                 | Prometheus/Grafana, ELK, alerting                                           | open source / hosted                                                                                                                                | build                                                                           |
| Integration                   | NHS Spine/Login/FHIR, DIATF register, GOV.UK                                | —                                                                                                                                                   | integrate                                                                       |

Because DIATF is now recognised in UK law (Data Act), being **DIATF-certified** is near-mandatory for an NHS identity service. You can be your own IDSP/HSP (certify via BSI/Kantara as already planned) **or** delegate proofing to certified providers (OneID/Yoti/Post Office) while you own the wallet+backend.

---

## 4. SUPPLIERS — DEEP SEARCH (shortlist, verified)

### *Card manufacturing & personalisation bureaus (UK/EU, certified secure)*

- **Certus Card Group** (UK) — end-to-end card+mailer+fulfilment, 46 yrs, contact/contactless, in-house encoding. Strong fit for pilot + ongoing.
- **Thales** (Discovery / IDemia) — global national-ID leader, end-to-end enrolment→issuance→derived mobile credentials. The benchmark for a national ID.
- **Entrust** — central + instant issuance, digital+physical, card personalisation; also HSMs (nShield) and PKI. Dual-role supplier.
- **G+D (Giesecke+Devrient)** — national-ID, secure cards, CMS, HSM expertise; major European gov-credential provider.
- **Global ID / HID Global** — secure card + identity programmes.

### *Chip / secure element*

- **Infineon** — SECORA ID (CC EAL6+, Java Card, dual-interface NFC) — ideal for a national ID chip.
- **NXP** — SmartMX / contactless NFC chips, very high volume.
- **IDemia (Thales)** — integrated.

### *HSMs / PKI*

- **Thales Luna / DPoD · Entrust nShield · Utimaco** (EU/UK-friendly).

### *CMS / issuance software*

- **Entrust** (ISS/central), **G+D**, **EastNets**, **Thales** — all proven at national scale.

### *Digital identity / identity wallet / proofing (DIATF-certified / UK)*

- **OneID®** — first DIATF v0.4 certified IDSP+OSP+HSP (Holder Service) in UK; a natural interoperability partner.
- **Post Office EasyID** — UK-certified IDSP/ISP.
- **Yoti** — IDSP, biometric, age/attr verification.
- **GOV.UK One Login** (GDS) — the government-level identity; NHS may expect alignment/interop.
- **ZEPHYR / Mitek, Onfido** — document/biometric proofing.

### ***Fulfilment & mail (UK)***

- **Certus** (mailers+fulfilment), **Royal Mail**, **Whistl**, **DocPost / Parity** — high-volume secure mail.

### ***Fraud / data***

- **CIFAS**, **LexisNexis Risk**, **Experian**, **Equifax**, **Onfido**.

### ***Card readers / hardware (enrolment & field)***

- **Zebra**, **HID Global**, **IDemia**, **ACS** (card readers).
- 

## **5. HEADCOUNT / ORG (indicative, to be "fully capable")**

| Function                                             | Staff                   |
|------------------------------------------------------|-------------------------|
| Programme leadership / NHS relationship              | 2–3                     |
| Programme/project managers + PMO                     | 4–6                     |
| Identity & PKI engineers (keys, HSMs, CMS)           | 4–6                     |
| Backend/app platform engineers (wallet, integration) | 8–12                    |
| Security (AEGIS ops + audits)                        | 4–6                     |
| Compliance / DP / DIATF / DSPT officer               | 2                       |
| Operations / fulfilment / helpdesk                   | 5–10 (scales w/ volume) |
| Fraud & risk analysts                                | 2–4                     |
| Supplier management / QA                             | 2–3                     |

**Rough core:** ~30–50 FTE to run autonomously; partners absorb the manufacturing/fulfilment volume.

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## **6. ESSENTIAL DELIVERABLES YOU MUST NOT SKIP (NHS/UK bar)**

**DIATF certification** (IDSP/HSP/Attribute SP) — via BSI or Kantara (already planned).

**GPG45 compliance** (identity proofing) — you or your proofing partner.

**GPG44 / passkey & PKI auth** — strong authentication.

**NHS app / Spine / FHIR integration.**

**ICO registration + DPIA** (ZC230615 in place).

**Data residency UK + DSPT/DTAC.**

**Card personalisation security:** card blank inventory control, secure printer/fulfilment chain, laminate, chip personalisation in HSM-signed environment.

**Physical card + digital wallet issuance pipeline** — order → personalise → activate → deliver.

**AEGIS woven through** every step (supply chain, enrolment, issuance, app).

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## 7. ROUGH COST SCALE (guide; confirm with bureau quotes)

| Item                                                   | Indicative range                                                                                      |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Program management + in-house platform build           | £3–10M                                                                                                |
| HSMs, PKI, secure infra                                | £0.5–2M                                                                                               |
| Physical card personalisation + materials (per card)   | £1.5–£5 per card (varies w/ chip + laminate + volume)                                                 |
| Fulfilment + mailer + postage (per card)               | £0.5–£2 per card                                                                                      |
| DIATF certification + security audits                  | £0.2–1M                                                                                               |
| Digital app/wallet + backend                           | £2–8M                                                                                                 |
| <b>Order of magnitude for 50M physical cards + app</b> | <b>£250M–£600M+</b> (dominant cost = cards + fulfilment + chip + postage at scale; not your software) |

This is why the realistic model = you own the **program/IP/software/AEGIS**, and the manufacturing/fulfilment is **contracted to certified bureaus at volume**.

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## 8. DELIVERY ROADMAP (suggested phases)

**Pilot** (months 1–6): in-house card printer + partner CMS/fulfilment; 1–10k cards; DIATF sandbox.

**Regional pilot** (6–12): bureau cards + mailer; regional issuance estate; HSM+ceremony live.

**Scale** (12–24): contracts with certified bureaus (Thales/G+D/Entrust/Certus), automated fulfilment, national app rollout, AEGIS at full load.

**National** (24+): 50M rollout plan with NHS; DHSC/NHS England commercial; DSPT/DTAC evidence.

**GATE:** NHS partnership route + DIATF certification + data residency + bureau contracts must all be green before scale.

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## 9. NEXT ACTION ITEMS (what I can do next)

- Draft **request-for-information (RFI)** emails to 3–4 bureaus (Certus, Entrust, Thales, G+D) for per-card pricing + capacity.
- Build the **10-page detailed architecture & supplier matrix** (the numbered capability owner for every layer).
- Draft **NHS partnership / commercial proposal** (this capability model + AEGIS + the £24.4bn/28x ROI case) for the 1-Oct call.
- Produce an **RFI spread** for fulfilment/mail + chip suppliers.

**END — All IP belongs to JDB Sales.**